



A more durable building material

A bio-inspired protein molecule additive - PEG-PVA, can help concrete resist freeze-damage. <http://digit.in/nov20-33>



Just an average day in Florida

A Florida resident's pet cat, dragged in a two-headed mutant snake as a gift of appreciation. <http://digit.in/nov20-34>

of tests, screening procedures and treatments for the previously incurable disease. Once the virus had been identified, it was possible to develop targeted antiviral drugs that could specifically attack the virus. Hepatitis C continues to affect many people around the world, but it is now possible to eradicate the disease just like smallpox or polio (which is still not considered to be totally eradicated). Millions of lives have been saved thanks to the efforts of these three scientists, as well as the chimpanzees.

PHYSICS

Half of the physics prize this year is awarded to Roger Penrose, with the other half shared between Reinhard Genzel and Andrea Ghez. Roger Penrose is considered to have made the biggest contribution to the general theory of relativity since Albert Einstein, with his 1965 paper that showed that the formation of black holes was a ramification of the general theory of relativity, that black holes could indeed form, and then went ahead to describe the properties of blackholes. The paper demonstrated that at the core of the black hole was a region of highly compressed spacetime that formed a singularity, within which all the known laws of physics broke down completely. This object had a gravitational influence so strong that even light could not escape. Penrose managed to prove something that Einstein himself did not believe in. To achieve this, Penrose attacked the theory of relativity with novel mathematics.

Reinhard Genzel and Andrea Ghez both studied the region near the centre of the Milky Way since the 1990s, by mapping the orbits of stars around an object now called Sagittarius A*. We now know that this object is a supermassive black hole, thanks to the work of these two scientists. Both the researchers came up with new techniques that allowed astronomers to study the object by peering through the disturbances of our own atmosphere, as well as the light, interstellar dust, and gas clouds that increase in

concentration towards the galactic centre. The scientists have been tracking the stars surrounding the supermassive black hole for over three decades, and they continue to improve on the instruments and techniques.

CHEMISTRY

Emmanuelle Charpentier and Jennifer A Doudna have been jointly awarded the Nobel Prize in chemistry. Clustered regularly interspaced short palindromic repeats or CRISPR are DNA sequences found in the immune systems of bacteria, that are made up of genetic material derived from bacteriophages or viruses that attacked the bacteria, and are used to detect and decimate similar DNA, which is effective at preventing future attacks. While studying a bacteria, *Streptococcus pyogenes*, Charpentier discovered a



Credit: Stephen McNally/UC Berkeley

Jennifer Doudna with a model of CRISPR-Cas9

previously unknown molecule called tracrRNA, which was part of the bacteria's CRISPR/Cas, which could cleave a DNA strand in two. Charpentier published the discovery in 2011, then collaborated with Doudna to modify and streamline the pair of scissors, and grow it in a test tube. The genetic scissors were made easier to use through this process. The team then proceeded to genetically modify the scissors, allowing them to cut through any arbitrary DNA molecule at a particular point in the chain, or site, even though the original DNA sequences in the bacteria were used specifically on viruses. In a 2014 podcast called Radiolab, Doudna describes the discovery in her own words as, "I really had the hairs on the back of my neck ... standing up,

just realizing this thing exists, and you could program it to cut DNA just like molecular scissors, and I can program it to cut DNA wherever I want." Since the discovery in 2012, the use of these genetic scissors has been taken up enthusiastically by scientists around the world, who now had the tools necessary to reprogram the molecule of life. There is plenty of ongoing work based on this one discovery, including developing new crops that are resistant to pests and mould, hardier crops that can withstand drought, new therapies that can treat cancer, as well as the possibility of stopping hereditary diseases from passing on to the next generation.

ECONOMICS

The oldest known examples of auctions occurred in Ancient Rome, where moneylenders recovered whatever they could by auctioning the items appropriated from the people who did not pay them back. Even in modern days, this practice continues by lenders, and it is not that auctions are restricted to fancy pieces in art circles. Auctions also take place regularly on the internet. In fact, the prices of a number of commodities are directly or indirectly determined by auction, including radio spectrum available for telecommunication, the amount of emissions a country can make, and the cost of electricity for the end user. These types of auctions have components that affect both private and public parties (emissions, electricity and spectrum), and therefore obtaining the maximum value for both the companies involved as well as the taxpayers is difficult to figure out. Paul R Milgrom and Robert B Wilson have studied how auctions work, why people behave in certain ways, and have developed new auction formats that are more equitable to everyone involved. These new auction formats are being used around the world today, including by official government regulators. 